

## Planes and lines in 3D

$\pi$  plane,  $A(x_0, y_0, z_0) \in \pi$

$\vec{v}(x_v, y_v, z_v), \vec{w}(x_w, y_w, z_w) \parallel \pi$

$\vec{n} = \{ \vec{v} \in V \mid \vec{v} \parallel \pi \} = \angle(\vec{v}, \vec{w})$

Parametric equations:  $\lambda, \mu \in \mathbb{R}$

$$\begin{cases} x = x_0 + \lambda x_v + \mu x_w \\ y = y_0 + \lambda y_v + \mu y_w \\ z = z_0 + \lambda z_v + \mu z_w \end{cases}$$

Cartesian equation (determinant point)

$$\begin{vmatrix} x - x_0 & y - y_0 & z - z_0 \\ x_v & y_v & z_v \\ x_w & y_w & z_w \end{vmatrix} = 0$$

Cartesian equation (implicit form)

$$A x + B y + C z + D = 0$$

$$\vec{n} = \langle \vec{v} \times \vec{w} \rangle = \langle A, B, C \rangle$$

$\ell$  line,  $A(x_0, y_0, z_0) \in \ell; \vec{v}(x_v, y_v, z_v) \parallel \ell$

Parametric equation

$$\begin{cases} x = x_0 + \lambda x_v \\ y = y_0 + \lambda y_v \\ z = z_0 + \lambda z_v \end{cases}$$

Cartesian equation (symmetric form)

$$\textcircled{1} \ x_v, y_v, z_v \neq 0$$

$$\frac{x - x_0}{x_v} = \frac{y - y_0}{y_v} = \frac{z - z_0}{z_v}$$

$$\textcircled{2} \ x_v = 0, y_v, z_v \neq 0$$

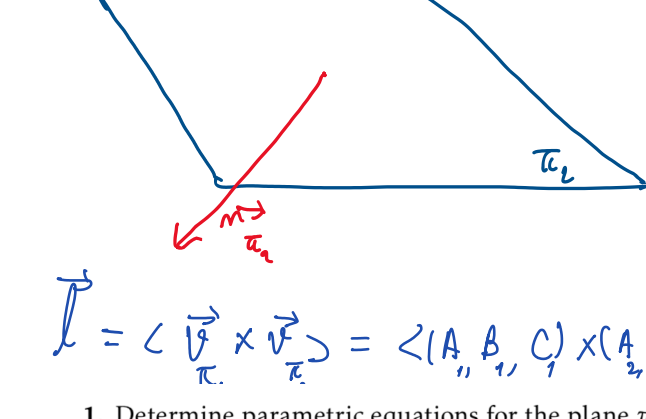
$$\begin{cases} x = x_0 \\ \frac{y - y_0}{y_v} = \frac{z - z_0}{z_v} \end{cases}$$

$$\textcircled{3} \ x_v = y_v = 0, z_v \neq 0$$

$$\begin{cases} x = x_0 \\ y = y_0 \end{cases}$$

Cartesian equation (implicit form)

$$\ell: \begin{cases} \pi_1: A_1 x + B_1 y + C_1 z + D_1 = 0 \\ \pi_2: A_2 x + B_2 y + C_2 z + D_2 = 0 \end{cases}$$



$$\vec{l} = \langle \vec{v}_{\pi_1} \times \vec{v}_{\pi_2} \rangle = \langle A_1, B_1, C_1 \rangle \times \langle A_2, B_2, C_2 \rangle$$

- Determine parametric equations for the plane  $\pi$  in the following cases:
  - $\pi$  contains the point  $M(1, 0, 2)$  and is parallel to the vectors  $\mathbf{a}_1(3, -1, 1)$  and  $\mathbf{a}_2(0, 3, 1)$ .
  - $\pi$  contains the points  $A(-2, 1, 1)$ ,  $B(0, 2, 3)$  and  $C(1, 0, -1)$ ,  $\rightarrow$  need 2 vectors
  - $\pi$  contains the point  $A(1, 2, 1)$  and is parallel to  $i$  and  $j$ ,  $\rightarrow$  axis
  - $\pi$  contains the point  $M(1, 7, 1)$  and is parallel coordinate plane  $Oy_2z_2$ ,  $\rightarrow$  axis
  - $\pi$  contains the points  $M_1(5, 3, 4)$  and  $M_2(1, 0, 1)$ , and is parallel to the vector  $\mathbf{a}(1, 3, -3)$ ,  $\rightarrow$  need additional vector  $\vec{v}_1, \vec{v}_2$
  - $\pi$  contains the point  $A(1, 5, 7)$  and the coordinate axis  $Ox_2$ ,  $\rightarrow$  axis

a) Cartesian equation

$$\begin{vmatrix} x - x_0 & y - y_0 & z - z_0 \\ x_{a_1} & y_{a_1} & z_{a_1} \\ x_{a_2} & y_{a_2} & z_{a_2} \end{vmatrix} = \begin{vmatrix} x - 1 & y & z - 2 \\ 3 & -1 & 1 \\ 0 & 3 & 1 \end{vmatrix} = 1 - x + 9z - 18 - 3x + 3 - 3y = 0$$

$$\begin{cases} x = 1 + 3\lambda \\ y = 0 - \lambda + 3\mu \\ z = 2 + \lambda + \mu \end{cases} \Rightarrow \begin{cases} x = 1 + 3\lambda \\ y = -\lambda + 3\mu \\ z = 2 + \lambda + \mu \end{cases}$$

- Determine Cartesian equations for the plane  $\pi$  in the following cases:

- $\pi: x = 2 + 3u - 4v, y = 4 - v, z = 2 + 3u$ ;
- $\pi: x = u + v, y = u - v, z = 5 + 6u - 4v$ .  $\rightarrow$   $\vec{v}_1(1, 1, 6), \vec{v}_2(1, -1, -4), \vec{n}(0, 0, 5)$

$$\pi: \begin{cases} x = 2 + \lambda \cdot 3 + \mu \cdot 4 \\ y = 4 - \lambda \cdot 0 + \mu \cdot (-1) \\ z = 2 + \lambda \cdot 3 + \mu \cdot 0 \end{cases} \Rightarrow \begin{vmatrix} x & y & z - 5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = 0$$

$$\vec{v}_1(1, 3, 0, 3), \vec{v}_2(-4, -1, 0) \Rightarrow \begin{vmatrix} x - 2 & y - 4 & z - 2 \\ -4 & -1 & 0 \\ 3 & 0 & 3 \end{vmatrix} = 0$$

- Determine parametric equations for the plane  $\pi$  in the following cases:

- $3x - 6y + z = 0$ ; (Dai valori ca parametru la  $x, y$ , sau  $z$ )
- $2x - y - z - 3 = 0$ ;

$$\text{a) } 3x - 6y + z = 0 \quad \text{b) } 2x - y - z - 3 = 0$$

$$\begin{cases} x = \lambda \\ y = \mu \\ z = 6\mu - 3\lambda \end{cases} \quad \begin{cases} x = \lambda \\ y = \mu \\ z = 2 - \mu - 3 \end{cases}$$

$$\vec{n} \in \langle (1, 0, -3), (0, 1, 6) \rangle \quad \vec{n} \in \langle (1, 0, 2), (0, 1, -1) \rangle$$

- Consider the lines  $\ell_1: x = 1 + t, y = 1 + 2t, z = 3 + t, t \in \mathbb{R}$  and  $\ell_2: x = 3 + s, y = 2s, z = -2 + s, s \in \mathbb{R}$ . Show that  $\ell_1$  and  $\ell_2$  are parallel and find the equation of the plane determined by the two lines.

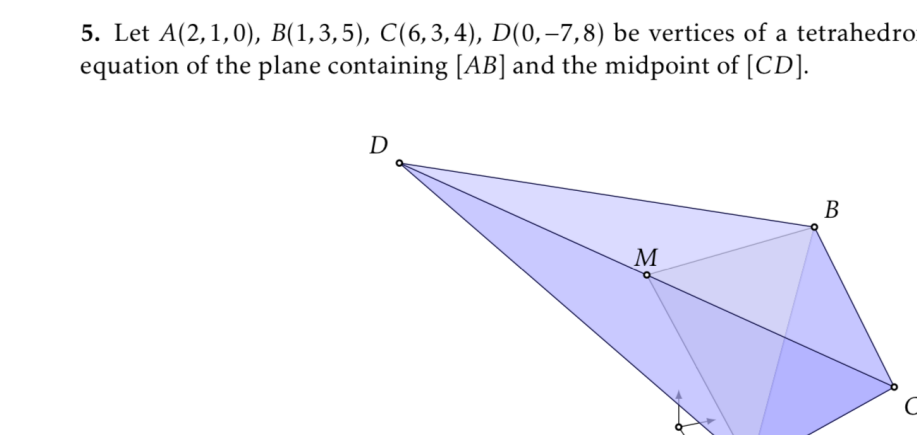
$$\ell_1: \begin{cases} x = 1 + t \\ y = 1 + 2t \\ z = 3 + t \end{cases} \Rightarrow \vec{t}(1, 2, 1)$$

$$\ell_2: \begin{cases} x = 3 + s \\ y = 2s \\ z = -2 + s \end{cases} \Rightarrow \vec{s}(1, 2, 1)$$

$$\ell_2: \begin{cases} x = 3 + s \\ y = 2s \\ z = -2 + s \end{cases} \Rightarrow \vec{s}(1, 2, 1)$$

$$\begin{cases} \vec{AB}(-2, -1, -5) \\ \vec{AC}(1, 2, 1) \\ \vec{AD}(1, 1, 3) \end{cases} \Rightarrow \text{Cartesian eq (Det form)} \begin{vmatrix} x - 1 & y - 1 & z - 3 \\ 2 & -1 & -5 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

- Let  $A(2, 1, 0)$ ,  $B(1, 3, 5)$ ,  $C(6, 3, 4)$ ,  $D(0, -7, 8)$  be vertices of a tetrahedron. Determine a Cartesian equation of the plane containing  $[AB]$  and the midpoint of  $[CD]$ .



$$\vec{n} \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right) = (3, -2, 6)$$

$$\vec{AB}(-1, 2, 5)$$

$$\vec{AM}(1, -1, 6)$$

- Show that a parallelepiped with faces in the planes  $2x + y - 2z + 6 = 0$ ,  $2x - 2y + z - 8 = 0$  and  $x + 2y + 2z + 1 = 0$  is rectangular.

$$\pi_1: 2x + y - 2z + 6 = 0$$

$$\pi_2: 2x - 2y + z - 8 = 0$$

$$\pi_3: x + 2y + 2z - 1 = 0$$

$$\vec{n}_1 \times \vec{n}_2 = (2, 1, -2) \times (2, -2, 1) = \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & -2 \\ 2 & 1 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & 1 \\ 2 & -2 \end{vmatrix} \vec{k} = -3\vec{i} - 6\vec{j} - 6\vec{k} = -3(1, 2, 2)$$

$$\vec{n}_1 \times \vec{n}_2 = \langle (-2, 1, 2) \rangle$$

$$\vec{n}_1 \times \vec{n}_2 = \langle (2, -2, 1) \rangle$$

Calculăm norma  $\Rightarrow \|\vec{n}_1 \times \vec{n}_2\| = \|\vec{n}_1\| \cdot \|\vec{n}_2\| = \|\vec{n}_1\| \cdot \|\vec{n}_2\| \Rightarrow$  ade

- Show that the points  $A(1, 0, -1)$ ,  $B(0, 2, 3)$ ,  $C(-2, 1, 1)$  and  $D(4, 2, 3)$  are coplanar.

$$\begin{vmatrix} x_0 - x_A & y_0 - y_A & z_0 - z_A \\ x_0 - x_B & y_0 - y_B & z_0 - z_B \\ x_0 - x_C & y_0 - y_C & z_0 - z_C \end{vmatrix} = 0 \Leftrightarrow A, B, C, D \text{ coplanar}$$

- Determine a Cartesian equation of the plane  $\pi$  if  $A(1, -1, 3)$  is the orthogonal projection of the origin on  $\pi$ .

$$\begin{cases} OA = \sqrt{11} \\ d(O, \pi) = \frac{|d|}{\sqrt{a^2 + b^2 + c^2}} \Rightarrow \frac{|d|}{\sqrt{a^2 + b^2 + c^2}} = \sqrt{11} \Rightarrow |d| = 11 \\ \vec{OA}(1, -1, 3) \Rightarrow \begin{cases} x = 2 \\ y = -2 \\ z = 3 \end{cases} \end{cases}$$

$$\ell \in \langle (1, -1, 3) \rangle$$

$$\pi: x - y + 3z + 11 = 0$$

- Determine the distance between the planes  $x - 2y - 2z + 7 = 0$  and  $2x - 4y - 4z + 17 = 0$ .

$$A(-2, 0, 0) \in \pi_1 \Rightarrow d(A, \pi_2) = d(A, \pi_1)$$

$$d(A, \pi_1) = \frac{|2(-2) - 0 + 7|}{\sqrt{2^2 + 4^2 + 4^2}} = \frac{3}{6} = \frac{1}{2}$$

- Determine the relative positions of the planes in the following cases

- $\pi_1: x + 2y + 3z - 1 = 0, \pi_2: x + 2y - 3z - 1 = 0$ .
- $\pi_1: x + 2y + 3z - 1 = 0, \pi_2: 2x + y + 3z - 2 = 0, \pi_3: x + 2y + 3z + 2 = 0$ .

$$\text{c) } \begin{cases} x + 2y + 3z - 1 = 0 \\ x + 2y - 3z - 1 = 0 \end{cases}$$

$$A: \begin{vmatrix} 1 & 2 & 3 & -1 \\ 1 & 2 & -3 & -1 \end{vmatrix} \xrightarrow{R_2 - R_1} \begin{vmatrix} 1 & 2 & 3 & -1 \\ 0 & 0 & 6 & 0 \end{vmatrix} \Rightarrow \text{rank } A = 2 \Rightarrow \text{the planes intersect}$$

$$\begin{cases} x + 2y + 3z - 1 = 0 \\ x + 2y - 3z - 1 = 0 \end{cases} \Leftrightarrow \begin{cases} x + 2y + 3z - 1 = 0 \\ z = 0 \end{cases} \Leftrightarrow \begin{cases} x + 2y = 1 \\ z = 0 \end{cases}$$

$$\text{Eq of the line: } \begin{cases} x + 2y = 0 \\ z = 0 \end{cases}$$

$$\text{d) } \pi_1 \text{ and } \pi_2 \text{ are parallel}$$

$$\begin{cases} x + 2y + 3z - 1 = 0 \\ 2x + y + 3z - 2 = 0 \end{cases} \Leftrightarrow \begin{cases} -x + y = 1 \\ 2 = y - \frac{1}{3} \end{cases} \quad ? ?$$

- Show that the planes

$$\pi_1: 3x + y + z - 1 = 0, \quad \pi_2: 2x + y + 3z + 2 = 0, \quad \pi_3: -x + 2y + z + 4 = 0$$

have a point in common.

$$\exists A(x, y, z) \in \pi_1 \text{ det } \neq 0$$

$$\begin{vmatrix} 3 & 1 & 1 \\ 2 & 1 & 3 \\ -1 & 2 & 1 \end{vmatrix} = -15 \neq 0$$

- Show that the pairwise intersection of the planes

$$\pi_1: 3x + y + z - 5 = 0, \quad \pi_2: 2x + y + 3z + 2 = 0, \quad \pi_3: 5x + 2y + 4z + 1 = 0$$

are parallel lines.

$$\pi_1: 3x + y + z - 5 = 0 \Rightarrow \vec{\ell}_1 \in \langle (3, 1, 1) \rangle$$

$$\pi_2: 2x + y + 3z + 2 = 0 \Rightarrow \vec{\ell}_2 \in \langle (2, 1, 3) \rangle$$

$$(3, 1, 1) \times (2, 1, 3) = \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} \vec{i} + \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} \vec{j} + \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} \vec{k} = 7\vec{i} - 7\vec{j} + 1\vec{k} \Rightarrow \langle (1, -1, 1) \rangle$$

Carte punct comun

$$\begin{cases} x = 0 \\ y + z = 5 \\ y + 3z = -2 \end{cases} \Leftrightarrow \begin{cases} z = \frac{-2}{2} \\ y = \frac{12}{2} \end{cases}$$

$$\begin{cases} x = 1 \\ y = -\frac{3}{2} - 3x \\ z = \frac{12}{2} + x \end{cases}$$

- Determine parametric equations for the line  $\ell$  in the following cases:

- $\ell$  contains the point  $M_0(2, 0, 3)$  and is parallel to the vector  $\mathbf{a}(3, -2, -2)$ ,
- $\ell$  contains the point  $A(1, 2, 3)$  and is parallel to the  $Oz$ -axis,
- $\ell$  contains the points  $M_1(1, 2, 3)$  and  $M_2(4, 4, 4)$ .

$$\text{a) } \frac{2x - 2}{3} = \frac{y}{-2} = \frac{z - 3}{-2} = t \Rightarrow \begin{cases} x = 2 + 3t \\ y = -2t \\ z = 3 - 2t \end{cases}$$

$$\begin{cases} \frac{x - 2}{3} = \frac{y}{-2} \\ \frac{y}{-2} = \frac{z - 3}{-2} \end{cases} \Leftrightarrow \begin{cases} 2x + 3y - 4 = 0 \\ y - z + 3 = 0 \end{cases}$$

$$\text{b) } \ell \parallel Oz \Rightarrow B(0, 0, z)$$

$$\begin{cases} x = 1 \\ y = 2 \end{cases}$$